$\#b$ nodes with a given in-degree

$n_b(K_b; t+1) = n(K_b, t)$

$+\ m \cdot \cancel{(1-g)} \cdot \Pi_b(K_b-1, t) \cdot n(K_b-1, t) \Longleftrightarrow$ (# promotions opportunities available · prob of promotion assigned for this level of skill & type of person · #ppl)

$-\ m \cdot \cancel{(1-g)} \cdot \Pi_b(K_b, t) \cdot n(K_b, t)$ (1)

$+\ (1-g)\delta_{K_b,0}$ (in BA new node has $K_a^{(in)} = K_b^{(in)} = 0$)

normalised, weighted expected value ansatz

$\text{prob}(\text{"b" node recieves new in-degree from new node}) = \Pi_{b,i} = \frac{(1-g)hK_{b,i} + g(1-h)K_{b,i} + \mu_b}{Z_b(t)} = \frac{\lambda_b K_{b,i} + \mu_b}{Z_b(t)}$

$\sum_{i \in U}^{N} (\Pi_{b,i} + \Pi_{a,i}) = 1$ as new node certainly gives some in-degree when $m > 0$

$\Rightarrow \sum_{i \in U_b \subset U}^{N_b} \Pi_{b,i} + \sum_{i \in U_a \subset U_b}^{N_a} \Pi_{a,i} = 1$ (*)

($\Leftrightarrow$ prob("b" node get in-degree) + prob("a" node gets in-degree) = 1)

$\lambda_b \sum_{i \in U_b}^{N_b} K_{b,i} = \lambda_b E_b(t)$

$\mu_b \sum_{i \in U_b}^{N_b} 1 = \mu_b N_b(t)$

for simplicity let: $N(t_{init}) = t_{init}$ & $E^{tot,in}(t_{init}) = m t_{init} = E_a^{in} + E_b^{in}$

lets assume $E_b^{in}(init) = m\,(1-g)ht + g(1-h)t_{init} = m\lambda_b t_{init}$

$N_b(t) + N_a(t) = N(t)$ $N_b(t) = (1-g)N(t)$

& that Since linear?

$\Rightarrow E_b(t) = \lambda_b m t$

& $N_b(t) = (1-g)t$

symmetric arg for "a":

$\Rightarrow E_a(t) = m[gh + (1-g)(1-h)]t = m\lambda_a t$

$\Rightarrow (*) \rightarrow \frac{(m\lambda_b^2 + (1-g)\mu_b)t}{Z_b(t)} + \frac{(m\lambda_a^2 + g\mu_a)t}{Z_a(t)} = 1$

The point of Z_a, Z_b is to ensure (*)

if we choose $Z_a = Z_b$ then the result is still true (?)

$\Rightarrow Z(t) = [m(\lambda_a^2 + \lambda_b^2) + (g\mu_a + (1-g)\mu_b)]t$

It's useful here to motivate $m\cancel{(1-g)}[m(\lambda_a^2 + \lambda_b^2) + (g\mu_a + (1-g)\mu_b)]^{-1} = \tilde{Z}$

$\Rightarrow (1) \rightarrow \therefore n(k_b, t+1) = n(k_b, t)$

$+\ \tilde{Z}\, t^{-1}(\lambda_b(K_b-1) + \mu_b)\, n(K_b-1, t)$

$-\ \tilde{Z}\, t^{-1}(\lambda_b K_b + \mu_b)\, n(K_b, t)$ (2)

$+\ (1-g)\delta_{K_{in},0}$

$(\lambda_a + \lambda_b = hg + (1-h)(1-g) + h(1-g) + (1-h)g$

$= (2hg + 1 - h - g) + (h + g - 2hg)$

$= 1)$, nice sanity check

$\Rightarrow E^{tot,in} = E^a + E^b$

$\lambda_a^2 + \lambda_b^2 = $ horrible ...